Final Assignment – FinTech Course

The final assignment requires students to address a given problem starting from the assigned dataset.

The delivery must include a well-structured code implementation (in a Jupyter Notebook or Google Colab Notebook). The code should be clearly commented, or alternatively accompanied by a written report describing the methodology, implementation, and results.

In addition, students are required to prepare a PowerPoint presentation that frames the problem, explains the approach taken, highlights the main results, and concludes with key insights and considerations.

Your assignment:

Bank Marketing (Dataset 4)

The data is related wit direct marketing campaigns of a Portuguese banking institution. The marketing campaigns were based on phone calls. Often, more than one contact to the same client was required, in order to access if the product (bank term deposit) would be (or not) subscribed.

Input variables:

bank client data:

1 - age (numeric)
2 - job : type of job (categorical:
"admin.","unknown","unemployed","management","housemaid","entrepreneur","student",
"blue-collar","self-employed","retired","technician","services")
3 - marital : marital status (categorical: "married","divorced","single"; note: "divorced" means divorced or widowed)
4 - education (categorical: "unknown","secondary","primary","tertiary")
5 - default: has credit in default? (binary: "yes","no")
6 - balance: average yearly balance, in euros (numeric)
7 - housing: has housing loan? (binary: "yes","no")
8 - loan: has personal loan? (binary: "yes","no")

related with the last contact of the current campaign:

9 - contact: contact communication type (categorical: "unknown","telephone","cellular")
10 - day: last contact day of the month (numeric)
11 - month: last contact month of year (categorical: "jan", "feb", "mar", …, "nov", "dec")
12 - duration: last contact duration, in seconds (numeric)

other attributes:

13 - campaign: number of contacts performed during this campaign and for this client (numeric, includes last contact)
14 - pdays: number of days that passed by after the client was last contacted from a previous campaign (numeric, -1 means client was not previously contacted)
15 - previous: number of contacts performed before this campaign and for this client (numeric)
16 - poutcome: outcome of the previous marketing campaign (categorical:
"unknown","other","failure","success")

Output variable (desired target):

17 - y - has the client subscribed a term deposit? (binary: "yes","no")

- *Create two different ML algorithms to compute the probability that a client will subscribe a term deposit, and compare them;*

- *Propose a strategy to decrease the number of features, and study the impact of this feature selection on the predictions.*
- *Provide interpretability results for the preferred algorithm (and explain why it is the “preferred” one).*